

AI-SA-IN-TH-000012

Module 8.1 · LLM Evaluation Fundamentals · AI Safety: Evaluation, Guardrails & Responsible Deployment

Fairness in AI Systems

Demographic Parity, Equalized Odds, Impossibility Theorem, and COMPAS Analysis

9 min

Duration

B3 – Apply

Bloom's

L3 – Advanced

Level

Theory

Domain

Learning Objectives

- 1 Define three statistical fairness metrics and write their formal mathematical definitions
- 2 Compute demographic parity and equalized odds gaps from a confusion matrix
- 3 State the fairness impossibility theorem and explain which real-world AI systems illustrate it
- 4 Apply the COMPAS recidivism case to identify conflicting fairness metrics in a deployed AI system
- 5 Explain why fairness definitions are context-dependent and select the appropriate metric for a given domain
- 6 Map AI fairness requirements to the applicable regulatory standards (EEOC, GDPR, EU AI Act)

What Is Fairness in AI?

Algorithmic Fairness

The absence of systematic, unjustifiable discrimination in AI-driven decisions — where 'discrimination' is defined in terms of outcomes across protected demographic groups or individuals with similar qualifications. Fairness is not a single property: multiple formal definitions exist, and choosing between them requires ethical and legal judgment.

- No single universal definition: the AI/ML literature contains 20+ distinct mathematical fairness definitions — they are not equivalent
- Legal grounding: algorithmic fairness intersects with anti-discrimination law; different legal standards (disparate impact, disparate treatment) map to different mathematical definitions
- Context-dependence: what constitutes a 'fair' AI system in a medical screening context is different from a credit scoring or recidivism prediction context
- Fairness is not the same as accuracy: a highly accurate model can be deeply unfair, and a fair model may sacrifice accuracy
- Process fairness vs outcome fairness: was the decision-making process unbiased? AND were outcomes equitable? Both must be addressed

Five Fairness Concepts — Visual Map

Demographic Parity

Equal positive outcome rate across groups

$$P(\hat{Y}=1 | A=0) = P(\hat{Y}=1 | A=1)$$

Equalized Odds

Equal TPR and FPR across groups

$$TPR_0 = TPR_1 \text{ and } FPR_0 = FPR_1$$

Predictive Parity

Equal precision (PPV) across groups

$$\frac{P(Y=1 | \hat{Y}=1, A=0)}{P(Y=1 | \hat{Y}=1, A=1)} = 1$$

Individual Fairness

Similar individuals get similar outcomes

$$|f(x_1) - f(x_2)| \leq d(x_1, x_2)$$

Counterfactual Fairness

Outcome same if only protected attr changed

$$P(\hat{Y} | \text{do}(A=a)) = P(\hat{Y} | \text{do}(A=a'))$$

Demographic Parity — Formal Definition & Worked Calculation

$$\text{Demographic Parity Gap} = |P(\hat{Y}=1 | A=0) - P(\hat{Y}=1 | A=1)|$$

1. Scenario: a loan approval AI; $A=0$ (White applicants), $A=1$ (Black applicants)
2. White applicants: 400 applicants, 280 approved $\rightarrow P(\hat{Y}=1 | A=0) = 280/400 = 0.70$
3. Black applicants: 200 applicants, 100 approved $\rightarrow P(\hat{Y}=1 | A=1) = 100/200 = 0.50$
4. Demographic Parity Gap = $|0.70 - 0.50| = 0.20$ (20 percentage points)
5. Interpretation: Black applicants are approved at a 20pp lower rate — a significant disparate impact violation
6. Threshold: gap ≤ 0.05 (5pp) is considered acceptable in most regulatory contexts; above 0.10 triggers audit

Note: Demographic parity ignores whether applicants are creditworthy — achieving parity may require approving more high-risk applicants in one group. Legal standard (ECOA) requires parity in creditworthy applicants, not all applicants.

Equalized Odds — Formal Definition & Worked Calculation

$$\text{Equalized Odds: } \text{TPR}_0 = \text{TPR}_1 \quad \text{AND} \quad \text{FPR}_0 = \text{FPR}_1$$

1. Scenario: same loan AI; now split by actual credit outcome ($Y=1$ = good credit, $Y=0$ = bad credit)
2. White: $\text{TP}=240$, $\text{FP}=40$, $\text{FN}=30$, $\text{TN}=90 \rightarrow \text{TPR}=240/270=0.889$, $\text{FPR}=40/130=0.308$
3. Black: $\text{TP}=80$, $\text{FP}=20$, $\text{FN}=40$, $\text{TN}=60 \rightarrow \text{TPR}=80/120=0.667$, $\text{FPR}=20/80=0.250$
4. $\text{TPR gap} = |0.889 - 0.667| = 0.222 \rightarrow$ creditworthy Black applicants approved at 22pp lower rate (false negative disparity)
5. $\text{FPR gap} = |0.308 - 0.250| = 0.058 \rightarrow$ non-creditworthy White applicants approved at 5.8pp higher rate
6. Both gaps violate equalized odds; the FPR gap means White applicants who should be rejected are approved at higher rates

Note: Equalized odds is stronger than demographic parity: it requires fairness conditional on the true outcome — controlling for actual creditworthiness. Preferred for high-stakes decisions.

Predictive Parity (Calibration)

Predictive Parity

A fairness criterion requiring that the positive predictive value (precision) is equal across groups: among all applicants the model predicts will repay the loan, the true repayment rate should be the same regardless of race, gender, or other protected attribute.

- Formal: $P(Y=1 \mid \hat{Y}=1, A=0) = P(Y=1 \mid \hat{Y}=1, A=1)$
- Intuition: when the model says 'this person will repay', that prediction should be equally reliable for all groups
- Also called calibration: a well-calibrated model's confidence scores reflect true probabilities equally across groups
- Conflict with equalized odds: the COMPAS recidivism algorithm satisfied predictive parity but violated equalized odds — this is the most famous fairness conflict in the field (Kleinberg et al., 2016; Chouldechova, 2017)
- When to use: when the key fairness concern is 'are high-risk predictions equally reliable' — common in credit scoring and risk assessment

Three Fairness Metrics — Properties, Tradeoffs, and When to Use

Metric	Controls For	Key Property	Best Suited For	Limitation
Demographic Parity	Nothing — raw rates	Equal selection rate across groups	Representation goals; anti-discrimination screening	Ignores whether selection is deserved; conflicts with meritocracy
Equalized Odds	True outcome	Equal TPR + FPR across groups	High-stakes: credit, hiring, healthcare	Hard to achieve simultaneously for both TPR and FPR
Predictive Parity	Positive predictions	Equal precision across groups	Risk scoring; resource allocation	May coexist with large disparities in FNR across groups

The Fairness Impossibility Theorem

Fairness Impossibility Theorem (Chouldechova 2017; Kleinberg et al. 2016)

When the base rates of the target variable differ between groups — which is typical in most real-world datasets — it is mathematically impossible for a binary classifier to simultaneously satisfy demographic parity, equalized odds (calibration), AND predictive parity. At most two of the three can be satisfied at once.

- Formal result: for groups with different base rates $p_0 \neq p_1$, satisfying any two of the three metrics forces a violation of the third
- Real-world implication: there is no 'fair' algorithm in an absolute sense — every choice involves an ethical tradeoff between competing definitions
- COMPAS example: the algorithm satisfied predictive parity (calibration) but violated equalized odds — ProPublica and Northpointe both cited correct statistics; they were measuring different things
- Design implication: before building a fair AI system, explicitly decide which fairness metric to prioritise — and document why in the model card
- Historical note: this theorem fundamentally changed the AI fairness discourse — the goal shifted from 'make a fair algorithm' to 'make explicit, justified fairness tradeoffs'

Group Fairness vs Individual Fairness

Group Fairness

- Measures equality of outcomes or error rates at the population level across demographic groups
- Legally well-grounded: maps directly to anti-discrimination law (disparate impact)
- Practical to measure: only requires group labels and outcome data
- Weakness: two individuals with identical qualifications may receive different treatment — group statistics can mask individual injustice
- Example: achieving demographic parity in hiring may require the model to apply different thresholds for different groups

Individual Fairness

- Measures equality of treatment for individuals with similar qualifications
- Intuitive fairness: the same person should get the same outcome regardless of which group they belong to
- Challenge: requires a task-appropriate 'similarity' metric $d(x_1, x_2)$ — defining this is an unsolved problem in most domains
- Weakness: cannot detect systemic discrimination invisible at the individual level
- Example: a fair hiring system treats Michael and Mohammed identically given identical CVs

COMPAS Recidivism Tool — ProPublica Analysis (2016)

Case: Correctional Offender Management Profiling for Alternative Sanctions (COMPAS)

- COMPAS: a commercial risk assessment tool by Northpointe used in US courts to predict likelihood of re-arrest, informing bail, sentencing, and parole decisions
- ProPublica investigation (2016): analysed 7,000+ defendants in Broward County, Florida using two years of follow-up data
- Key finding 1: Black defendants labelled 'high risk' but did not re-offend at twice the rate of White defendants (false positive rate: Black 45%, White 23%)
- Key finding 2: White defendants labelled 'low risk' who did re-offend at higher rate than Black defendants labelled 'low risk' (false negative rate: White 48%, Black 28%)
- Northpointe's response: COMPAS is calibrated — among defendants scored 7/10 risk, the recidivism rate is the same regardless of race (predictive parity satisfied)

Key Lesson

Both ProPublica and Northpointe were mathematically correct — they measured different fairness metrics. COMPAS satisfies predictive parity but violates equalized odds. Since Black and White defendants have different base recidivism rates in the data, the impossibility theorem guarantees it cannot satisfy both. This case made the impossibility theorem widely known outside academia.

Fairness Metric Conflict in COMPAS — Worked Example

Demonstrating that predictive parity and equalized odds cannot both hold when base rates differ

COMPAS Approximate Statistics (Broward County)

Base recidivism rate: Black defendants = 51%, White defendants = 39%

Among 'High Risk' predictions:

Black: PPV = 63% (calibrated)

White: PPV = 59% (calibrated — approx. equal → predictive parity ≈ satisfied)

Among all defendants:

False Positive Rate: Black = 45%, White = 23% → NOT equal

False Negative Rate: Black = 28%, White = 47% → NOT equal → equalized odds VIOLATED

Why Both Cannot Hold

If you force $FPR_{Black} = FPR_{White}$ (equalized odds), you must set different risk thresholds for each group. But different thresholds mean different PPV values (predictive parity broken).

The impossibility theorem in action: different base rates (51% vs 39%) make simultaneous satisfaction mathematically impossible.

Policy question: which error is more harmful — falsely labelling someone high-risk (FPR) or failing to predict someone who re-offends (FNR)? This is an ethical question, not a mathematical one.

Context-Dependence — Different Fairness Definitions for Different Domains

Medical Screening

Recommended metric:
Equalized Odds
(Equal TPR)

Missing a disease in one group
is more harmful than false alarms

Hiring / Lending

Recommended metric:
Demographic Parity
or Equalized Odds

Legal requirement: equal opportunity
+ consider EEOC / ECOA context

Criminal Justice

Recommended metric:
Equalized Odds
(Equal FPR)

False positive (false accusation)
is most severe harm; prioritise FPR

There is no universally correct fairness metric — the choice must be justified by domain context, harm severity, and applicable law.

Fairness by Domain — Medical vs Hiring vs Lending

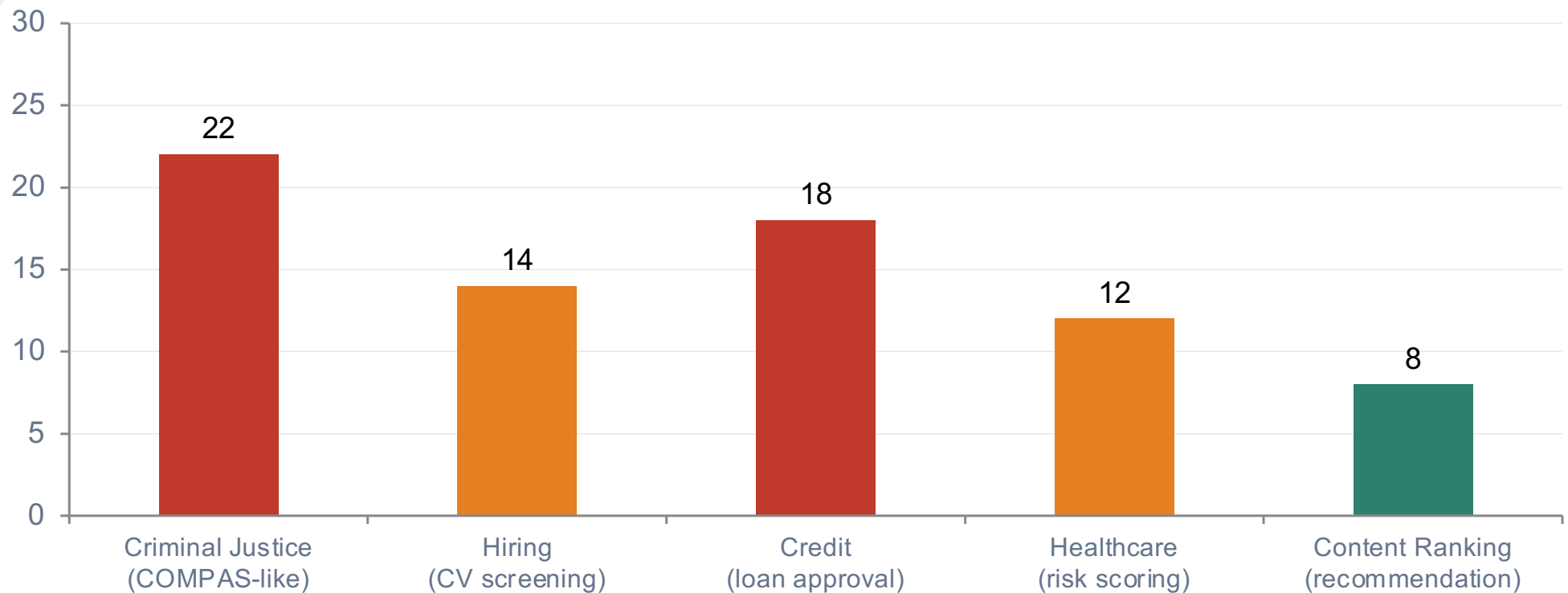
Domain	Harm of FP	Harm of FN	Prioritised Metric	Legal Standard
Medical screening	Unnecessary treatment (moderate)	Missed disease (severe)	High recall = equalise TPR	ADA; Section 1557 (ACA)
Hiring	Unqualified hire (moderate)	Rejected qualified candidate (mod.)	Demographic parity + equalized odds	EEOC; Title VII; EU Equal Treatment
Credit scoring	Approved bad loan (moderate)	Rejected creditworthy applicant (mod.)	Predictive parity + demographic parity	ECOA; Fair Housing Act; GDPR Art 22
Criminal justice	False conviction (very severe)	Missed repeat offender (moderate)	Equalise FPR	14th Amendment; due process; ECHR Art 6
Content ranking	Irrelevant content shown (low)	Good content suppressed (low-mod.)	Demographic parity in exposure	No specific standard yet — emerging

Regulatory Landscape — EEOC, Fair Housing, GDPR, EU AI Act

Key regulations governing algorithmic fairness in consequential AI decisions:

- EEOC Uniform Guidelines (1978, US): the 4/5ths (80%) adverse impact rule for employment selection procedures — applies to any selection test, including AI hiring tools
- Equal Credit Opportunity Act / Fair Housing Act (US): prohibit discrimination in credit and housing decisions; disparate impact is actionable; AI-driven credit scoring and property valuation tools are covered
- GDPR Article 22 (EU): individuals have the right not to be subject to decisions based solely on automated processing that significantly affect them — requires human review pathway and explanation on request
- EU AI Act (2024): 'high-risk AI systems' in employment, credit, education, and law enforcement require conformity assessment, bias testing, detailed documentation, and human oversight — must log protected attribute performance
- NYC Local Law 144 (2021): first US law requiring bias audits for AI hiring tools — annual third-party audits with public disclosure of impact ratios by sex and race/ethnicity required
- India DPDP Act (2023): includes provisions for accuracy and non-discrimination in automated processing of personal data — emerging fairness obligations for AI deployed in India

Demographic Parity Gap Across Industry Sectors



Source: aggregated from published bias audit studies 2018–2024. Criminal justice and credit scoring show largest gaps. A gap > 10pp is generally considered unacceptable under most regulatory frameworks — all five sectors exceed this threshold in typical unaudited deployments.

Key Takeaways

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- 1 There is no universal definition of fairness — 20+ mathematical definitions exist; demographic parity, equalized odds, and predictive parity are the three most widely used
- 2 The fairness impossibility theorem proves that when base rates differ between groups, no binary classifier can simultaneously satisfy all three metrics — choosing between them is an ethical decision, not a technical one
- 3 COMPAS satisfied predictive parity but violated equalized odds — ProPublica and Northpointe were both mathematically correct, measuring different things — illustrating why the choice of metric must be made explicit
- 4 Fairness definitions are context-dependent: medical screening should prioritise equal TPR; criminal justice should prioritise equal FPR; each domain's harm structure determines the appropriate metric
- 5 Regulations are catching up: NYC Local Law 144 requires annual AI hiring bias audits with public disclosure; EU AI Act mandates bias testing for high-risk AI; GDPR Art 22 requires human review for automated high-stakes decisions